PG No. 12 D_4 422 [tetragonal]

* Wyckoff site: `2a`, site symmetry: `4..`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `4..`

No.	vector	center	mapping
1	[0, 0, Z]	[0, 0, z]	`[1,2,3,4]`
2	[0, 0, $-Z$]	[0, 0, $-z$]	`[5,6,7,8]`

Table 2: Wyckoff bond: `4b@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	[X, Y, 0]	[0, 0, z]	`[1,-2]`
2	[$-Y$, X, 0]	[0, 0, z]	`[3,-4]`
3	[$-X$, Y, 0]	[0, 0, $-z$]	`[5,-6]`
4	[Y, X, 0]	[0, 0, $-z$]	`[7,-8]`

Table 3: Wyckoff bond: `8c@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	[X, Y, Z]	[0, 0, z]	`[1]`
2	[$-X$, $-Y$, Z]	[0, 0, z]	`[2]`
3	[$-Y$, X, Z]	[0, 0, z]	`[3]`
4	[Y, $-X$, Z]	[0, 0, z]	`[4]`
5	[$-X$, Y, $-Z$]	[0, 0, $-z$]	`[5]`
6	[X, $-Y$, $-Z$]	[0, 0, $-z$]	`[6]`
7	[Y, X, $-Z$]	[0, 0, $-z$]	`[7]`
8	[$-Y$, $-X$, $-Z$]	[0, 0, $-z$]	`[8]`

* Wyckoff site: `4b`, site symmetry: `..2`

Table 4: Wyckoff bond: `4a@4b`, bond symmetry: `..2`

No.	vector	center	mapping
1	[X, $-X$, Z]	[x, x, 0]	`[1,-7]`
2	[$-X$, X, Z]	[$-x$, $-x$, 0]	`[2,-8]`
3	[X, X, Z]	[$-x$, x, 0]	`[3,-5]`
4	[$-X$, $-X$, Z]	[x, $-x$, 0]	`[4,-6]`

Table 5: Wyckoff bond: `4b@4b`, bond symmetry: `..2`

No.	vector	center	mapping
1	[X, X, 0]	[x, x, 0]	`[1,7]`
2	[$-X$, $-X$, 0]	[$-x$, $-x$, 0]	`[2,8]`
3	[$-X$, X, 0]	[$-x$, x, 0]	`[3,5]`
4	[X, $-X$, 0]	[x, $-x$, 0]	`[4,6]`

Table 6: Wyckoff bond: `8c@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -x, 0]$	`[2]`
3	$[-Y, X, Z]$	$[-x, x, 0]$	`[3]`
4	$[Y, -X, Z]$	$[x, -x, 0]$	`[4]`
5	$[-X, Y, -Z]$	$[-x, x, 0]$	`[5]`
6	$[X, -Y, -Z]$	$[x, -x, 0]$	`[6]`
7	$[Y, X, -Z]$	$[x, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[-x, -x, 0]$	`[8]`

* Wyckoff site: `4c`, site symmetry: `.2.`

Table 7: Wyckoff bond: `4a@4c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[x, 0, 0]$	`[1,-6]`
2	$[0, -X, Z]$	$[-x, 0, 0]$	`[2,-5]`
3	$[-X, 0, Z]$	$[0, x, 0]$	`[3,-7]`
4	$[X, 0, Z]$	$[0, -x, 0]$	`[4,-8]`

Table 8: Wyckoff bond: `4b@4c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,6]`
2	$[-X, 0, 0]$	$[-x, 0, 0]$	`[2,5]`
3	$[0, X, 0]$	$[0, x, 0]$	`[3,7]`
4	$[0, -X, 0]$	$[0, -x, 0]$	`[4,8]`

Table 9: Wyckoff bond: `8c@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, 0, 0]$	`[2]`
3	$[-Y, X, Z]$	$[0, x, 0]$	`[3]`
4	$[Y, -X, Z]$	$[0, -x, 0]$	`[4]`
5	$[-X, Y, -Z]$	$[-x, 0, 0]$	`[5]`
6	$[X, -Y, -Z]$	$[x, 0, 0]$	`[6]`
7	$[Y, X, -Z]$	$[0, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, -x, 0]$	`[8]`

* Wyckoff site: `8d`, site symmetry: `1`

Table 10: Wyckoff bond: `8a@8d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$[-x, -y, z]$	`[2]`
3	$[-Y, X, Z]$	$[-y, x, z]$	`[3]`
4	$[Y, -X, Z]$	$[y, -x, z]$	`[4]`
5	$[-X, Y, -Z]$	$[-x, y, -z]$	`[5]`
6	$[X, -Y, -Z]$	$[x, -y, -z]$	`[6]`
7	$[Y, X, -Z]$	$[y, x, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$[-y, -x, -z]$	`[8]`